

The empirical recurrence for OEIS A235003: recovering a conjecture that is not written down

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Abstract

OEIS A235003 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 392$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235003 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 2, with no adjacent elements equal (constant-stress tilted 1×1 tilings)”. It has offset 1 and begins

1946, 14182, 114796, 1049470, 10000516, 98790336, 988422984, 9992949442, ...

The entry states:

Empirical recurrence of order 70 (see link above).

As of the “Last modified” line on the live entry (Jun 20 2022 00:49:52, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 392$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 392$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 784$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	17	c_{19}	−1849037535858108	c_{37}	231177171578558305262	c_{55}	56638285570293
c_2	161	c_{20}	4305240577836871	c_{38}	−1567652028613041455173	c_{56}	41672940675754
c_3	−3753	c_{21}	16322557046437122	c_{39}	−218376899842097867014	c_{57}	−1532066933025
c_4	−9776	c_{22}	−40434532170880778	c_{40}	2439978288324980164592	c_{58}	−814555790789
c_5	379286	c_{23}	−116057027438396084	c_{41}	60464137997297910976	c_{59}	3003011759879
c_6	161862	c_{24}	305787939425981877	c_{42}	−3120595502419467662814	c_{60}	1151783075158
c_7	−23325770	c_{25}	667056191170626911	c_{43}	216933397513372560316	c_{61}	−410129755555
c_8	13984213	c_{26}	−1876553356190194768	c_{44}	3267604542555923801377	c_{62}	−113024686464
c_9	978902121	c_{27}	−3103982798290107303	c_{45}	−480704736453599455390	c_{63}	367024836373
c_{10}	−1155605882	c_{28}	9398543533969386347	c_{46}	−2786914114996958625329	c_{64}	728738555412
c_{11}	−29801110650	c_{29}	11683344693281143710	c_{47}	587112218701432304006	c_{65}	−19404105782
c_{12}	46222106468	c_{30}	−38579972117677996163	c_{48}	1922658598908236649308	c_{66}	−2856589514
c_{13}	683134927671	c_{31}	−35427003193926815536	c_{49}	−504042531359578663966	c_{67}	49344456970
c_{14}	−1237214004952	c_{32}	130192903654288856027	c_{50}	−1063197124415984566060	c_{68}	5829883228
c_{15}	−12083759329066	c_{33}	85745718425349406223	c_{51}	322140471169118779966	c_{69}	−320784157
c_{16}	24294030642460	c_{34}	−361909541130650529567	c_{52}	465761584784964398621	c_{70}	−34867843
c_{17}	167738139484976	c_{35}	−162522811166523815225	c_{53}	−155589433732881793736		
c_{18}	−365173907355114	c_{36}	829518730435188059390	c_{54}	−159230820266097953359		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235003.
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